

Database Management System (DBMS)

Introduction

A **Database Management System (DBMS)** is software that allows users to create, store, organize, retrieve, and manage data efficiently. It acts as an interface between the user and the database, ensuring that data remains secure, consistent, and easily accessible. DBMS reduces redundancy, improves data integrity, and supports multiple users accessing the same database simultaneously.

Objectives of DBMS

- Store large amounts of data efficiently.
 - Retrieve data quickly using queries.
 - Reduce data redundancy.
 - Maintain data consistency and integrity.
 - Provide data security and access control.
 - Support concurrent access by multiple users.
 - Facilitate backup and recovery.
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Features of DBMS

- Data abstraction
 - Data independence
 - Data security
 - Data integrity
 - Concurrency control
 - Backup and recovery
 - Multiple user support
 - Query processing using SQL
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Advantages of DBMS

- Eliminates duplicate data.
 - Improves data consistency.
 - Provides better data security.
 - Enables data sharing among multiple users.
 - Simplifies data management.
 - Supports backup and recovery.
 - Ensures efficient data retrieval.
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Disadvantages of DBMS

- High implementation cost.
- Requires skilled administrators.
- Complex software architecture.
- Performance overhead for small applications.
- Maintenance costs can be high.